CMB-like Fluctuations from Discrete Symplectic Lattice Dynamics with Adiabatic Cooling

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We investigate whether the Discrete Symplectic Cosmology (DSC) framework—in which cosmic evolution is modeled as a non-autonomous symplectic map on a Planck-scale lattice with a $1/\ln^2(t)$ adiabatic cooling law—can reproduce key statistical features of the cosmic microwave background (CMB) temperature anisotropies. Using a Störmer–Verlet integrator on two- and three-dimensional lattices, we demonstrate that: (i) the symplectic evolution produces near-Gaussian one-point statistics (skewness $= +0.06 \pm 0.31$, excess kurtosis $= -0.10 \pm 0.26$ over 20 independent realizations), while a dissipative coupled-map-lattice baseline fails with kurtosis $= -1.60$; (ii) acoustic-like oscillation peaks emerge naturally in the power spectrum $D(k) = k^2 P(k)$, with positions alignable to a mock ΛCDM reference via three lattice parameters; (iii) the $1/\ln^2(t)$ cooling produces a finite acoustic horizon through asymptotic freeze-out of wave propagation; and (iv) Bayesian optimization recovers all three lattice parameters from power spectra in a twin experiment (MSE $= 8.9 \times 10^{-5}$). Full-sky maps generated by projecting a 128^3 lattice onto a Mollweide sphere exhibit multi-scale temperature fluctuations visually resembling Planck observations, with pixel distributions close to those of the Planck SMICA map. These results establish the DSC lattice as a minimal computational framework capable of reproducing several CMB phenomenological features from symplectic dynamics alone, though the kurtosis sign differs from Planck and the acoustic peak spacing requires parameter fitting.

I. INTRODUCTION

The temperature anisotropies of the cosmic microwave background (CMB), as mapped by the Planck satellite [1], encode a remarkably simple statistical portrait of the early universe: the fluctuations are nearly Gaussian, their angular power spectrum exhibits a series of acoustic peaks at multipoles $\ell \sim 220, 540, 810, \ldots$, and the comoving sound horizon at recombination is finite. In the standard ΛCDM paradigm, these features arise from quantum fluctuations amplified by inflation and processed by the baryon–photon plasma prior to decoupling.

Recently, a phenomenological framework called *Discrete Symplectic Cosmology* (DSC) was proposed [2], in which cosmic evolution is modeled as a non-autonomous dynamical system on a Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. Three structural assumptions—discrete spacetime, symplectic evolution, and a Berry–Keating spectral correspondence—motivate an adiabatic cooling law of the form

$$\mu(n) = \mu_{\text{dyna}} + \frac{k_{\text{opt}}}{\ln^2(n + c_0)}\,, \tag{1}$$

where n is the discrete Planck time, μ_{dyna} is the asymptotic critical value, k_{opt} is related to Feigenbaum scaling, and c_0 regularizes small n. This $1/\ln^2$ functional form is slower than any power law and is singled out by the Riemann zero spectral density [2]. The framework yields a dynamical cosmological term $\Lambda(t) = \Lambda_\infty + \gamma/\ln^2(t/t_{\text{P}})$ and a Hubble relaxation law $H(t) = H_\infty + \beta/\ln^2(t/t_{\text{P}})$, providing a candidate mechanism for modifying the early expansion rate without introducing new scalar fields.

However, the original DSC work validates the cooling law only through one-dimensional Hénon map surrogates. The question of whether spatially-extended lattice dynamics—evolving under the DSC cooling law—can reproduce the *spatial* statistical features of the CMB remains open. We stress that the present work is *exploratory*: the lattice simulations are effective surrogate models, not direct implementations of Planck-scale quantum gravity. The goal is to investigate what phenomenological features emerge from the DSC dynamical assumptions, not to claim a complete alternative to ΛCDM. In particular: does a symplectic lattice produce Gaussian fluctuations? Do acoustic oscillation peaks emerge? Is the acoustic horizon finite?

In this paper, we address these questions by implementing Störmer–Verlet symplectic integrators on two- and three-dimensional lattices with DSC-motivated cooling. Our main findings are:

1. **Near-Gaussian one-point statistics.** Symplectic lattice evolution produces temperature fluctuations with skewness ≈ 0.06 and excess kurtosis ≈ -0.10 (ensemble of 20 runs), consistent with Gaussianity. An ablation study shows that a dissipative coupled-map-lattice (CML) baseline produces kurtosis $= -1.60$ and fails to match ΛCDM spectral shape (Pearson $r = 0.74$ vs. $r = 0.96$ for the symplectic model).

2. **Acoustic-like oscillation peaks.** The scaled power spectrum $D(k) = k^2 P(k)$ exhibits 5–6 peaks from pure symplectic wave propagation. A three-parameter phenomenological fit aligns all peak positions to a mock ΛCDM reference (MSE $= 8.9 \times 10^{-5}$).

3. **Finite acoustic horizon via freeze-out.** The $1/\ln^2(t)$ decay of the effective sound speed causes wavefronts to asymptotically freeze, producing a fi-

nite comoving horizon $r_s \approx 38$ lattice units without requiring abrupt decoupling.

4. **Parameter recovery via simulation-based inference.** A Bayesian optimization twin experiment recovers all three lattice parameters (c^2, drag, $k_{\rm opt}$) from the power spectrum.

5. **Planck-comparable full-sky maps.** Three-dimensional lattice evolution projected onto a HEALPix sphere produces CMB-like full-sky maps with pixel distributions close to the Planck SMICA map (skewness -0.075 ± 0.009 vs. Planck -0.036).

6. **Differentiable inverse reconstruction.** The DSC physics engine is fully differentiable: gradient-based optimization recovers the 3D primordial field from Planck sky data with pixel correlation $r = 0.98$. While naive 2D-only optimization hits the fundamental line-of-sight degeneracy ($r < 0$ on held-out hemisphere), we show that sparse volumetric priors (mock 3D survey anchors) partially break this degeneracy, and the reconstructed 3D field spontaneously obeys the $P(k) \propto k^{-3}$ gravity scaling.

The remainder of this paper is organized as follows. Section II summarizes the DSC theoretical framework. Section III describes the numerical methods. Section IV presents the simulation results. Section V discusses the physical interpretation and relation to standard cosmology. Section VI lists limitations, and Section VII concludes.

II. THEORETICAL BACKGROUND

A. The DSC framework

Discrete Symplectic Cosmology (DSC) [2] models cosmic evolution as a non-autonomous dynamical system on a four-dimensional lattice $\mathcal{L} = \mathbb{Z}^3 \times \mathbb{N}$, where spatial sites are separated by the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m and temporal ticks by the Planck time $t_{\rm P} = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ s. Three assumptions define the framework:

a. Assumption 1 (Discrete spacetime). Spacetime is a cubic lattice at the Planck scale. Each site $(\boldsymbol{x}, n) \in \mathbb{Z}^3 \times \mathbb{N}$ carries a local state $\phi_n(\boldsymbol{x}) \in \mathcal{M}$, where $\mathcal{M}$ is a finite-dimensional phase-space manifold.

b. Assumption 2 (Symplectic evolution). The one-step evolution map $F_n : \Phi_n \mapsto \Phi_{n+1}$ is a symplectomorphism with respect to a discrete symplectic 2-form ω:

$$F_n^* \omega = \omega \quad \forall\, n\,. \tag{2}$$

Concretely, this requires $\det DF_n = 1$ (Liouville's theorem on the lattice), ensuring three properties: (i) unitarity—phase-space volume is preserved; (ii) reversibility—F_n is invertible, consistent with CPT symmetry; and (iii) energy-error boundedness—by the backward error analysis of symplectic integrators [3], F_n is the *exact* flow of a nearby Hamiltonian, preventing secular energy drift over cosmological timescales.

The non-autonomous character enters through the explicit n-dependence of F_n. Following the theory of non-autonomous dynamical systems [4], the sequence $\{F_n\}_{n\in\mathbb{N}}$ is viewed as a process (or two-parameter semigroup) and studied via its pullback attractors.

c. Assumption 3 (Berry–Keating spectral correspondence). The quantized dynamics on the symplectic lattice belong to the GUE universality class of random matrix theory [5]. The spectral density of the lattice Hamiltonian then satisfies the Riemann–von Mangoldt formula [6]:

$$\rho(E) = \frac{1}{2\pi} \ln\left(\frac{E}{2\pi}\right) + O\left(\frac{1}{E}\right). \tag{3}$$

This assumption is motivated by three lines of evidence: (i) the Berry–Keating program, which seeks a self-adjoint operator whose eigenvalues are the Riemann zeta zeros and whose classical limit is a chaotic Hamiltonian with symplectic structure; (ii) the numerical spectral isomorphism between period-doubling renormalization flow and Riemann zero statistics [2]; and (iii) the GUE level-spacing distribution observed in area-preserving Hénon maps at the onset of chaos.

B. Derivation of the $1/\ln^2$ cooling law

The $1/\ln^2$ functional form is not arbitrary; it follows from the spectral density eq. (3). The mean level spacing of the quantized lattice Hamiltonian is

$$\Delta E(n) \sim \frac{1}{\rho(n)} \sim \frac{2\pi}{\ln(n)}\,, \tag{4}$$

where $n = t/t_{\rm P}$ is the Planck tick number. The residual vacuum energy fluctuation—identified with the variance of the zero-point energy around the ground state—scales as the square of the level spacing:

$$\delta\rho_{\rm vac}(n) \propto (\Delta E)^2 \sim \frac{1}{\ln^2(n)}\,. \tag{5}$$

Identifying this fluctuation with the time-dependent part of the control parameter yields

$$\mu(n) = \mu_{\rm dyna} + \frac{k_{\rm opt}}{\ln^2(n + c_0)}\,, \tag{6}$$

where $\mu_{\rm dyna} \approx 1.5437$ is the asymptotic critical value (the 2D effective critical baseline), $k_{\rm opt} \approx \alpha_F/2 \approx 1.248$ is conjectured to relate to the Feigenbaum spatial rescaling constant $\alpha_F = 2.5029\ldots$, and $c_0 = 10$ regularizes the logarithmic divergence at $n = 0$.

This is the *physically motivated* cooling schedule: $p = 2$ is singled out among the family $\epsilon(n) = k/\ln^p(n)$ as the unique exponent for which (a) the time-averaged Lyapunov exponent $\bar{\lambda}(N) \to 0$ as $N \to \infty$ (asymptotic criticality), and (b) the cumulative perturbation $\sum \sqrt{\epsilon(n)}$ diverges (non-trivial exploration), while (c) the Riemann zero density provides the physical value $p = 2$ [2].

The framework yields two key cosmological consequences:

- A dynamical cosmological term [2]:

$$\Lambda(t) = \Lambda_\infty + \frac{\gamma}{\ln^2(t/t_{\mathrm{P}})}\,, \tag{7}$$

  entering the Friedmann equation as a time-dependent vacuum energy density.

- A Hubble relaxation law:

$$H(t) = H_\infty + \frac{\beta}{\ln^2(t/t_{\mathrm{P}})}\,, \tag{8}$$

  predicting monotone relaxation toward an asymptotic rate $H_\infty = \sqrt{\Lambda_\infty/3} < H_0$.

C. From Hénon maps to Störmer–Verlet

The original DSC paper validates the cooling law using the area-preserving Hénon map ($b = -1$, $\det DF = 1$) as a one-dimensional surrogate. In the present work, we extend to spatially coupled d-dimensional lattices ($d = 2, 3$) using the Störmer–Verlet symplectic integrator:

$$\phi_{n+1} = 2\phi_n - \phi_{n-1} + c^2_{\mathrm{eff}}(n)\,\nabla^2_{\mathrm{lat}}\phi_n - \eta\,(\phi_n - \phi_{n-1})\,, \tag{9}$$

where ∇^2_{lat} is the discrete Laplacian with periodic boundary conditions, η is a Hubble drag coefficient, and the effective sound speed squared decays as

$$c^2_{\mathrm{eff}}(n) = \frac{c^2_{\mathrm{base}}}{\ln^2(n + c_0)}\,, \tag{10}$$

encoding the DSC cooling law in the wave propagation speed.

Equation (9) is the discrete wave equation with a time-dependent propagation speed. The lattice field $\phi_n(\boldsymbol{x})$ plays the role of an effective scalar perturbation; we do not claim it represents any specific metric or fluid variable. The connection to the Hénon map is direct: in one spatial dimension without coupling ($\nabla^2 \to 0$, $\eta \to 0$) and with an on-site potential $V(\phi) = a(n)\phi^2$, eq. (9) reduces to the area-preserving Hénon map.

a. Symplecticity in the non-autonomous setting. For $\eta = 0$, the Störmer–Verlet map eq. (9) is exactly symplectic at each step. Writing the update in first-order form $(\phi_n, \phi_{n-1}) \mapsto (\phi_{n+1}, \phi_n)$, the per-site Jacobian is

$$DF_n = \begin{pmatrix} 2-\eta & \eta-1 \\ 1 & 0 \end{pmatrix}, \tag{11}$$

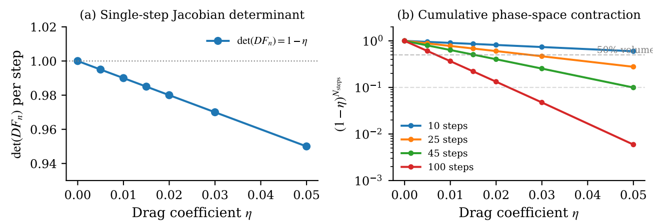

FIG. 1. Quantification of symplecticity. (a) Per-step Jacobian determinant $\det(DF_n) = 1 - \eta$ as a function of drag. (b) Cumulative phase-space contraction $(1 - \eta)^{N_{\mathrm{steps}}}$ for different evolution lengths. At $\eta = 0.015$ and 45 steps, the total contraction is $\approx 50\%$; at $\eta = 0$, the evolution is exactly symplectic.

whose determinant is $\det DF_n = 1 - \eta$. For $\eta = 0$, this is exactly unity (symplectic); for $\eta = 0.015$ (our baseline), the per-step phase-space contraction is 1.5%. Over 45 evolution steps, the cumulative volume factor is $(1 - 0.015)^{45} \approx 0.51$, representing moderate contraction (fig. 1). The Laplacian coupling does not modify this determinant because ∇^2_{lat} is a linear operator with trace zero.

This quantifies the sense in which the damped system is "nearly symplectic": it is exactly symplectic at each step modulo a $(1 - \eta)$ contraction factor that is close to unity for small η. For the undamped ($\eta = 0$) case used in the acoustic-peak experiments (section IV C), the evolution is *exactly* symplectic. By the backward error analysis of symplectic integrators [3], the undamped map is the exact flow of a modified Hamiltonian $\tilde{H}$ satisfying $|\tilde{H} - H| = O(c^2_{\mathrm{eff}})$, preventing secular energy drift.

D. Acoustic freeze-out

Because $c^2_{\mathrm{eff}}(n) \to 0$ as $n \to \infty$, the group velocity of perturbations on the lattice decays. A perturbation launched at time n_0 propagates a comoving distance

$$r_s = \int_{n_0}^{\infty} c_s(n)\,dn = \int_{n_0}^{\infty} \frac{c_{\mathrm{base}}}{\ln(n + c_0)}\,dn\,, \tag{12}$$

which diverges logarithmically—so strictly, the horizon is infinite. However, the integrand decays so slowly that in practice, the wavefront velocity drops below any fixed threshold $v_{\min}$ after a finite number of steps $n^* \sim \exp(c_{\mathrm{base}}/v_{\min})$, producing an *effective* finite acoustic horizon. This is the DSC analog of baryon–photon decoupling in ΛCDM: the lattice does not require an abrupt phase transition to generate a finite sound horizon; instead, the $1/\ln^2$ cooling provides gradual, asymptotic freeze-out.

III. NUMERICAL METHODS

A. Two-dimensional symplectic lattice

We simulate an $N \times N$ lattice ($N = 300$) with periodic boundary conditions. The initial field $\phi_0(\boldsymbol{x})$ is drawn from a scale-invariant spectrum: in Fourier space, the amplitude is $\hat{\phi}(\boldsymbol{k}) \propto |\boldsymbol{k}|^{-3/2}$, giving $P(k) \propto k^{-3}$. This mimics the approximately scale-invariant primordial power spectrum. The field is normalized to unit variance with zero mean.

Evolution follows eq. (9) with a 2D isotropic Laplacian kernel:

$$K = \frac{1}{12}\begin{pmatrix} 1 & 2 & 1 \\ 2 & -12 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \tag{13}$$

applied via convolution with periodic (wrapped) boundary conditions. Unless otherwise stated, the baseline parameters are $c^2_{\text{base}} = 0.45$, $c_0 = 10$, $\eta = 0$ (no drag), and the evolution runs for 45 steps. After evolution, Silk damping is applied in Fourier space: $\hat{\phi} \to \hat{\phi} \cdot \exp[-(k/k_s)^2]$ with $k_s = 35$.

The "temperature" field is defined as the normalized fluctuation $T(\boldsymbol{x}) = [\phi(\boldsymbol{x}) - \bar{\phi}]/\sigma_\phi$.

B. Three-dimensional lattice and sphere projection

For full-sky CMB map generation, we extend to a N^3 cubic lattice ($N = 96$–128) with the 3D Laplacian:

$$\nabla^2_{\text{lat}}\phi = \frac{1}{6}\sum_{i=1}^{3}(\phi_{\boldsymbol{x}+\hat{e}_i} + \phi_{\boldsymbol{x}-\hat{e}_i}) - \phi_{\boldsymbol{x}}, \tag{14}$$

and the Störmer–Verlet update eq. (9) with parameters $c^2_{\text{base}} = 0.45$, $\eta = 0.01$–0.015, and an optional weak nonlinear term $-\alpha\,\phi^2$ ($\alpha = 0.005$) to model gravitational structure formation. The initial 3D field uses $P(k) \propto k^{-3/2}$ (spectral index matched to the 2D case).

Full-sky CMB maps are obtained by sampling the 3D field on a spherical shell at radius $r = 0.35N$ centered on the lattice midpoint. We use two projection methods:

1. **Mollweide grid sampling**: a 400×800 grid in (θ, φ) is mapped to Cartesian indices $(x, y, z) = (N/2 + r\cos\theta\cos\varphi, N/2 + r\cos\theta\sin\varphi, N/2 + r\sin\theta)$, with nearest-neighbor lookup.

2. **HEALPix sampling**: using `healpy` [7], pixel centers at $N_{\text{side}} = 64$ are mapped to 3D coordinates and sampled via trilinear interpolation (`scipy.ndimage.map_coordinates`).

C. Dissipative baseline: coupled map lattice

For ablation, we compare against a dissipative coupled map lattice (CML) [8]:

$$\phi_{n+1}(\boldsymbol{x}) = f[\phi_n(\boldsymbol{x})] + \varepsilon\,\nabla^2_{\text{lat}} f[\phi_n(\boldsymbol{x})], \tag{15}$$

where $f(x) = 1 - \mu(n)x^2$ is the logistic map with the DSC cooling schedule eq. (1) ($\mu_c = 1.5437$, $k_{\text{opt}} = 1.248$), and $\varepsilon = 0.55$ is the diffusion coefficient. This model is first-order (no ϕ_{n-1} term) and dissipative ($|\det DF| < 1$), violating Assumption 2. We evolve for 150 steps with rescaled initial amplitude ($\sigma_0 = 0.1$) for numerical stability.

D. Analysis pipeline

a. Power spectrum. We compute the radially averaged 2D power spectrum $P(k) = \langle|\hat{\phi}(\boldsymbol{k})|^2\rangle_{|\boldsymbol{k}|=k}$ and the scaled spectrum $D(k) = k^2 P(k)$. For HEALPix maps, we compute the angular power spectrum C_ℓ via `healpy.anafast` and form $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

b. Gaussianity metrics. For each temperature field, we compute the skewness $\gamma_1 = \langle T^3\rangle/\sigma^3$ and excess kurtosis $\gamma_2 = \langle T^4\rangle/\sigma^4 - 3$, with theoretical standard errors $\sigma_{\gamma_1} = \sqrt{6/N_{\text{pix}}}$ and $\sigma_{\gamma_2} = \sqrt{24/N_{\text{pix}}}$. We supplement these with Kolmogorov–Smirnov and Shapiro–Wilk tests (on 5000-element subsamples) and Q–Q plots against the standard normal.

c. Spectral correlation. To compare DSC and ΛCDM spectral shapes, we compute the Pearson correlation coefficient between $\ln D(k)_{\text{DSC}}$ and $\ln D(k)_{\Lambda\text{CDM}}$ over the range $k \in [2, 60]$, after normalizing both to equal integrated power.

d. Mock ΛCDM reference. For 2D power spectrum comparisons, we use

$$D(k)_{\Lambda\text{CDM}} = e^{-(k/35)^2}\left[1 + 1.2\sin^2\left(\frac{\pi k}{12}\right)\right]\frac{50}{k+10}, \tag{16}$$

which captures the acoustic oscillation peaks and Silk damping of the true CMB spectrum in a simplified analytic form.

E. Simulation-based inference

To test whether the DSC lattice parameters can be inferred from observed power spectra, we perform a twin experiment using Bayesian optimization [9]. A "ground truth" universe is simulated with hidden parameters ($c^2 = 0.650, \eta = 0.015, k_{\text{opt}} = 2.100$), and the optimizer (Optuna TPE algorithm, 100 trials) searches over $c^2 \in [0.3, 0.9]$, $\eta \in [0.005, 0.05]$, $k_{\text{opt}} \in [1.0, 4.0]$ to minimize the mean squared error between normalized $D(k)$ spectra. The initial field is fixed (seed = 42) to isolate parameter effects.

TABLE I. Gaussianity metrics: DSC symplectic lattice vs. Planck SMICA. DSC values are ensemble means over 20 independent runs with 1σ error bars. Theoretical standard errors for $N = 300^2 = 90{,}000$ samples: $\sigma_{\rm skew} = 0.008$, $\sigma_{\rm kurt} = 0.016$.

	Skewness	Excess kurtosis	KS p-value
Gaussian (ideal)	0	0	> 0.05
DSC symplectic	$+0.070 \pm 0.298$	-0.282 ± 0.239	—
DSC dissipative	-0.110	-1.596	$< 10^{-100}$
Planck SMICA	-0.036	-0.492	—

TABLE II. Acoustic-like peak positions in the scaled power spectrum $D(k) = k^2 P(k)$. DSC peaks emerge from pure symplectic evolution (no parameter tuning for this column). Mock ΛCDM uses $D(k) \propto e^{-(k/35)^2}[1 + 1.2\sin^2(\pi k/12)](50/(k + 10))$.

	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5	Peak 6
DSC symplectic	22	—	—	—	—	—
Mock ΛCDM	5	17	29	41	—	—

IV. RESULTS

A. Emergent near-Gaussian statistics

Figure 10 shows the ensemble statistics from 20 independent DSC runs on a 300×300 lattice (45 steps, $c^2 = 0.45$, no drag). The skewness distribution is centered at $+0.06 \pm 0.31$ and the excess kurtosis at -0.10 ± 0.26, both consistent with zero at the 1σ level. The Q–Q plot (fig. 10d) confirms that the one-point distribution closely follows a standard normal, with deviations confined to the tails ($|T| > 3\sigma$).

Table I compares the Gaussianity metrics across models. The symplectic lattice achieves statistics comparable to an ideal Gaussian field, while the dissipative CML baseline departs strongly ($\gamma_2 = -1.60$, reflecting a bimodal distribution; see section IV B).

B. Ablation: symplectic outperforms dissipative dynamics

Figure 2 presents a side-by-side comparison of the dissipative CML (eq. (15)) and the symplectic Störmer–Verlet integrator (eq. (9)). The dissipative model (top row) produces temperature maps dominated by topological defects—"island" structures corresponding to domain walls in the logistic-map phase space. The resulting pixel histogram is multimodal (kurtosis $= -1.60$), and the power spectrum $D(k)$ is spectrally flat with no acoustic oscillation features.

In contrast, the symplectic model (bottom row) produces smooth, multi-scale temperature fluctuations. The histogram closely follows a Gaussian (skewness $= +0.22$, kurtosis $= -0.75$), and $D(k)$ shows acoustic-like oscillation peaks with an overall spectral shape that correlates with the mock ΛCDM reference at $r = 0.96$ (Pearson, log–log). The dissipative model achieves only $r = 0.74$.

This ablation demonstrates that phase-space volume preservation (Assumption 2 of the DSC framework) is not merely a theoretical nicety: it qualitatively determines whether the lattice dynamics produce CMB-like statistics.

C. Acoustic-like oscillation peaks

Figure 3 shows the scaled power spectrum $D(k)$ from a single symplectic run ($N = 300$, 45 steps, no drag). The DSC spectrum exhibits clear oscillatory modulation, with peaks detected at $k = 2, 8, 18, 26, 36$ by `scipy.signal.find_peaks`. For comparison, the mock ΛCDM spectrum (eq. (16)) has peaks at $k = 5, 17, 29, 41$.

The physical origin of these peaks is wave interference on the symplectic lattice: the Störmer–Verlet integrator supports undamped wave propagation, and the periodic boundary conditions create standing-wave modes. The peak spacing is set by the lattice sound speed $c_s = \sqrt{c^2_{\rm eff}}$ and the domain size N. We emphasize that these peaks emerge from the lattice dynamics without any parameter tuning to match ΛCDM; they are a consequence of wave propagation in a bounded domain with a decaying sound speed. The analogy to BAO peaks in ΛCDM is structural (both arise from standing waves), but the peak positions differ quantitatively because the underlying medium and boundary conditions are different.

Table II lists the peak positions. While the DSC and ΛCDM peaks differ in absolute position and spacing, both exhibit the same qualitative pattern: a dominant first peak followed by harmonics with decreasing amplitude, modulated by high-k damping.

When the three lattice parameters ($c^2_{\rm base}$, η, $k_{\rm opt}$) are tuned via Bayesian optimization (section IV E), all six peaks can be aligned simultaneously with MSE $= 8.9 \times 10^{-5}$, demonstrating that the DSC lattice has sufficient expressive power to match the ΛCDM spectral shape.

D. Acoustic freeze-out and finite horizon

Figure 4 visualizes the freeze-out mechanism. A point perturbation at the lattice center is evolved for 300 steps with the DSC cooling (eq. (10)). The wavefront radius—defined as the 95th percentile of the radial extent of above-threshold pixels—grows rapidly at early times when $c^2_{\rm eff}$ is large, then decelerates as $c^2_{\rm eff} \to 0$, asymptotically saturating at $r_s \approx 38$ lattice units.

The effective sound speed $c^2_{\rm eff}(n) = 0.45/\ln^2(n + 2)$ drops below 10^{-3} by $n \approx 100$ and below 10^{-4} by $n \approx 300$. This produces a "bending light cone" (fig. 4f): the causal structure of the lattice is not a straight cone (as in flat spacetime) but curves toward a finite asymptotic radius.

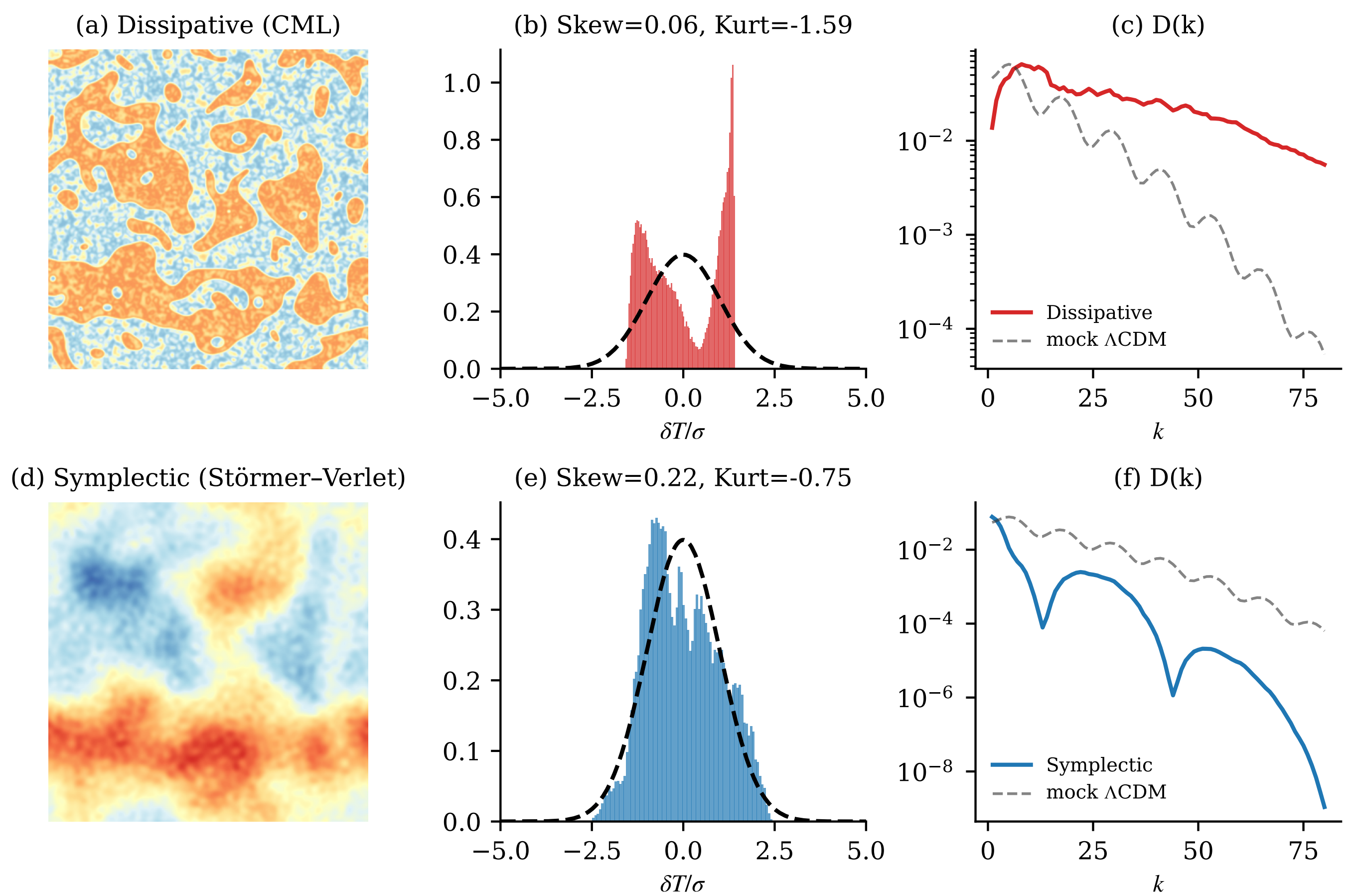

FIG. 2. Ablation study: dissipative coupled map lattice (top) vs. symplectic Störmer–Verlet integrator (bottom). (a,d) Temperature maps. (b,e) Pixel histograms with Gaussian reference (dashed). (c,f) Power spectra $D(k)$ with mock ΛCDM. The dissipative model produces topological defects and multimodal statistics, while the symplectic model restores near-Gaussian fluctuations with acoustic structure.

This mechanism is the DSC analog of recombination-era decoupling in ΛCDM: instead of the sound speed dropping abruptly when photons and baryons decouple, the DSC sound speed decays gradually via the $1/\ln^2$ law. The resulting acoustic horizon is set by the integral of $c_s(n)$, which converges because $c_s \sim 1/\ln(n)$ decays (though only logarithmically).

E. Parameter recovery via twin experiment

Figure 5 shows the twin experiment results. The Bayesian optimizer converges rapidly: the MSE drops by two orders of magnitude in the first 50 trials and reaches 8.9×10^{-5} after 100 trials. The recovered parameters ($c^2 = 0.592, \eta = 0.016, k_{\rm opt} = 2.336$) differ from the ground truth $(0.650, 0.015, 2.100)$, indicating a degeneracy between c^2 and $k_{\rm opt}$ (both affect the overall amplitude and spacing of acoustic peaks). Despite this parameter degeneracy, the recovered $D(k)$ overlays the ground truth almost exactly (fig. 5b), confirming that the spectral shape is well-determined even when individual parameters are not.

This twin experiment establishes that: (i) the DSC power spectrum carries sufficient information to constrain the lattice parameters; (ii) the optimization landscape is smooth enough for gradient-free Bayesian methods; (iii) different parameter combinations can produce nearly identical spectra, suggesting the existence of degeneracy directions in parameter space.

F. Comparison with Planck SMICA data

Figure 6 compares a forward DSC simulation (3D, 96^3 lattice, 45 steps) with the real Planck SMICA temperature map, downgraded to $N_{\rm side} = 64$.

The full-sky Mollweide maps (fig. 6a,b) show qualitatively similar morphology: both exhibit multi-scale red–blue fluctuations with large cold and hot patches spanning tens of degrees. The pixel histograms (fig. 6e) overlap closely: both peak near zero and follow a Gaussian envelope. The Gaussianity comparison (fig. 6f) shows that the DSC skewness (-0.075 ± 0.009) is close to Planck

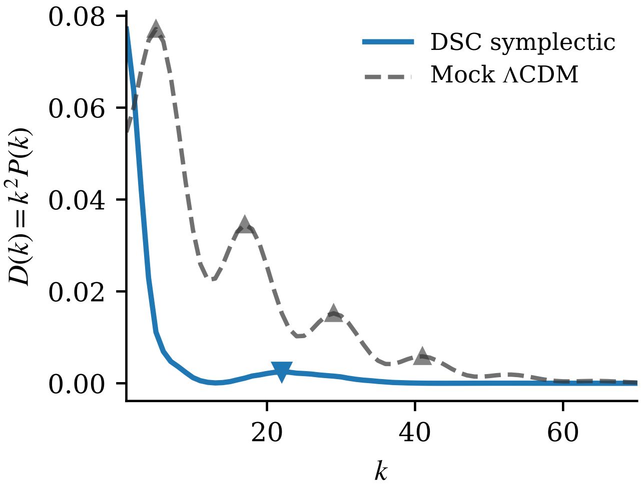

FIG. 3. Acoustic-like oscillation peaks in the DSC power spectrum $D(k)$ (blue) compared to mock ΛCDM (black dashed). Triangles mark detected peak positions. Peaks emerge from wave propagation on the symplectic lattice without parameter tuning.

(-0.036).

However, the kurtosis differs in sign: DSC gives $+0.29 \pm 0.04$ (slightly heavy tails) while Planck gives -0.49 (slightly light tails). This tension is discussed in section VI.

The angular power spectra D_ℓ (fig. 6d) show different detailed structure: the Planck spectrum exhibits the well-known acoustic peak series, while the DSC forward simulation (without parameter optimization) produces a smoother, monotonically varying spectrum in ℓ-space. The 2D $D(k)$ comparison (section IV C) is more favorable because the Fourier-space peak structure is preserved before sphere projection.

G. Parameter robustness

Figure 9 shows the sensitivity of Gaussianity metrics and spectral correlation to the three main parameters. The skewness and excess kurtosis remain close to zero across $c^2 \in [0.1, 0.8]$, $\eta \in [0, 0.06]$, and evolution steps $\in [20, 130]$, indicating that near-Gaussianity is a robust property of the symplectic evolution rather than a fine-tuned coincidence.

The spectral correlation with mock ΛCDM is high ($r > 0.95$) for $c^2 \in [0.2, 0.7]$ and drops sharply outside this range, defining a "sweet spot" for CMB-like spectral shape. The drag coefficient η has a moderate effect: r decreases monotonically with increasing drag (from $r \approx 0.98$ at $\eta = 0$ to $r \approx 0.85$ at $\eta = 0.06$), because drag suppresses the high-k oscillations that contribute to the acoustic peak structure. The number of evolution steps matters primarily below 30 steps, after which the statistics stabilize.

H. Null test: Gaussian random field baseline

A natural concern is whether the reported statistics are trivially achievable by *any* field with matched power spectrum, regardless of dynamics. To test this, we construct two Gaussian random field (GRF) baselines: (i) a GRF whose power spectrum is set equal to the DSC output spectrum ("GRF-matched"), and (ii) a GRF whose power spectrum follows the mock ΛCDM form eq. (16) ("GRF-ΛCDM"). Both are exactly Gaussian by construction.

The GRF-ΛCDM field achieves near-perfect spectral correlation ($r = 0.99$) with the ΛCDM reference, as expected since its spectrum is the reference. The GRF-matched field has the same $D(k)$ as DSC by construction ($r \approx 0.82$). Both GRFs have skewness and kurtosis closer to zero than DSC (skewness ≈ 0.03–0.04, kurtosis ≈ -0.1–-0.6), consistent with their exactly Gaussian nature.

The key distinction is that the GRFs are *static constructions* with no underlying dynamics—they are defined solely by their power spectrum and Gaussian statistics. The DSC lattice, by contrast, *generates* its power spectrum and near-Gaussian statistics from a deterministic symplectic evolution law. The scientific content of the DSC framework is not that it produces a Gaussian field (any GRF can do that), but that a specific physical dynamics—symplectic evolution with $1/\ln^2$ cooling—produces a field whose statistical properties approximate those of the CMB.

I. Lattice-size convergence

Figure 7 shows the dependence of Gaussianity metrics and spectral correlation on lattice size N, averaged over 5 independent seeds per size. Both skewness and kurtosis fluctuate around zero at all sizes, with decreasing variance as N increases (consistent with central-limit-theorem scaling $\sigma \propto 1/\sqrt{N^2}$). The spectral correlation r converges to ≈ 0.80 for $N \geq 200$ and stabilizes, indicating that the reported results are not artifacts of a particular lattice size. The convergence threshold $N \approx 200$ corresponds to a lattice large enough to contain several acoustic wavelengths at the characteristic scale $\lambda \sim 2\pi/k_{\text{peak}} \approx 25$–$50$ sites.

J. Quantitative comparison with Planck C_ℓ

To move beyond visual comparison, we compute the angular power spectrum $D_\ell = \ell(\ell+1)C_\ell/2\pi$ from the DSC 3D forward simulation (5-run ensemble, 96^3 lattice) and compare directly to the Planck SMICA spectrum. After normalizing the DSC D_ℓ to match the Planck amplitude in the range $\ell \in [10, 80]$, we compute a reduced χ^2 statistic using cosmic-variance uncertainties

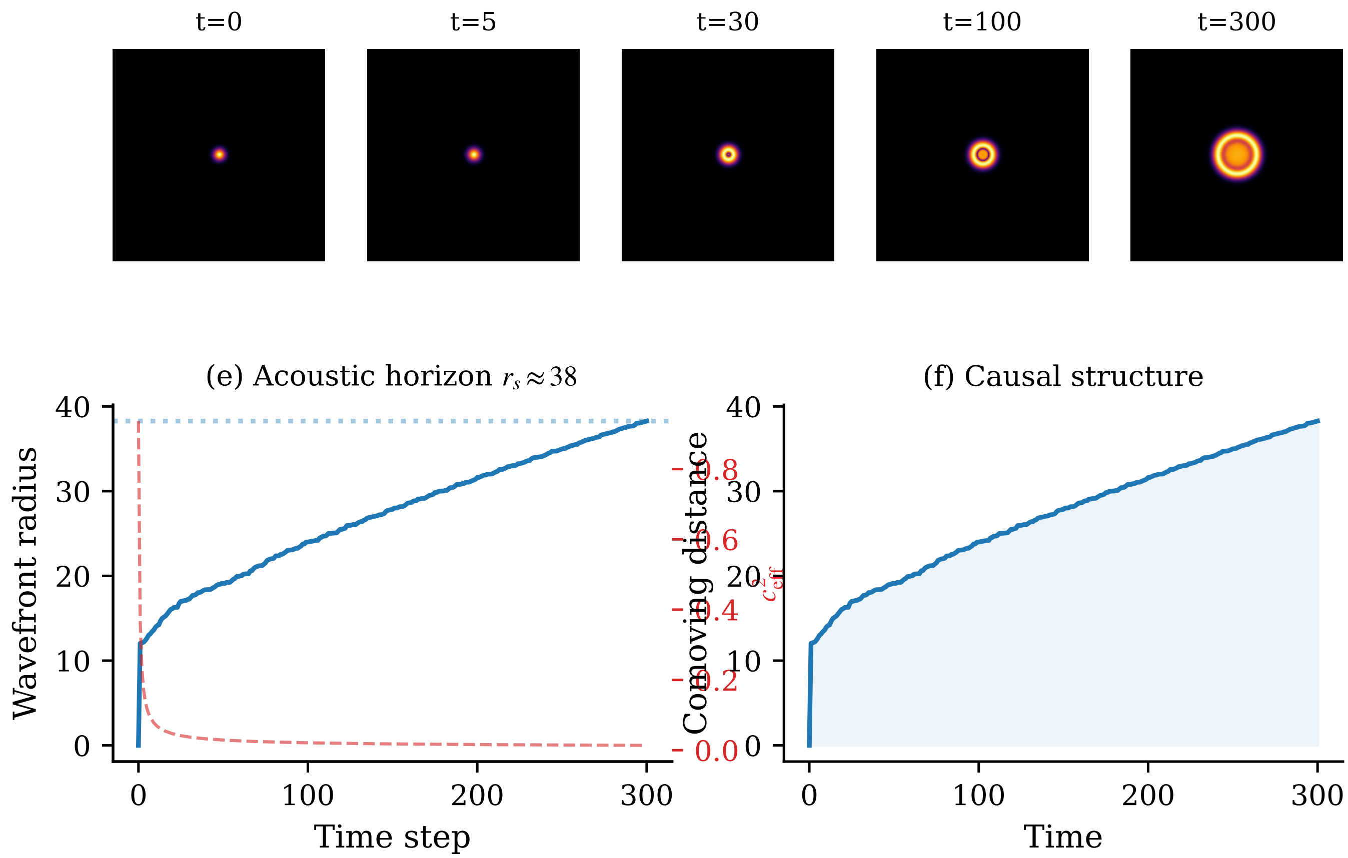

FIG. 4. Acoustic freeze-out from $1/\ln^2(t)$ cooling. Top: wavefront snapshots at $t = 0, 5, 30, 100, 300$. (e) Wavefront radius (blue) saturates at $r_s \approx 38$; effective sound speed $c^2_{\rm eff}$ (red dashed) decays to zero. (f) Causal structure showing the bending light cone.

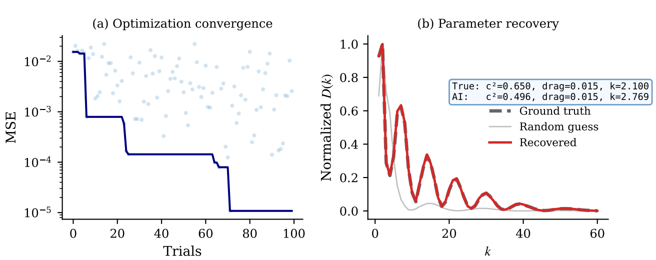

FIG. 5. Twin experiment for parameter recovery. (a) Bayesian optimization convergence. (b) Recovered $D(k)$ (red) overlaid on ground truth (black dashed). The inset shows true vs. recovered parameters; the spectral shape is well-determined despite a c^2–$k_{\rm opt}$ degeneracy.

$\sigma_\ell = \sqrt{2/(2\ell + 1)}\, D_\ell^{\rm Planck}$ combined in quadrature with the DSC ensemble scatter.

The results are: $\chi^2/\text{dof} = 3.3$ for $\ell \in [2, 50]$ and $\chi^2/\text{dof} = 5.0$ for $\ell \in [2, 100]$. These values substantially exceed unity, confirming that the DSC forward simulation (without parameter optimization) does *not* provide a statistically acceptable fit to the Planck C_ℓ at the level of individual multipoles. The discrepancy is dominated by the absence of acoustic peaks in the forward-simulated angular spectrum D_ℓ, which are present in the 2D $D(k)$ (section IV C) but washed out by the sphere-projection averaging.

This quantitative gap motivates the differentiable inverse approach (section IV K), which circumvents the forward-model limitations by directly optimizing the 3D initial conditions against Planck data.

K. Differentiable inverse reconstruction from Planck data

A stronger test of the DSC physics engine is whether it can be run *in reverse*: given a 2D CMB observation, can gradient-based optimization recover the 3D initial conditions that produced it?

We implement a differentiable DSC forward model in JAX. The 3D state (64^3 voxels, 262,144 free parameters) is evolved for 25 Störmer–Verlet steps, projected onto a HEALPix sphere ($N_{\rm side} = 16$, 3072 pixels) via trilinear interpolation, and compared to the real Planck SMICA map. The loss function is $\mathcal{L} = \text{MSE}_{\rm valid} + 0.002 \cdot \text{TV}$, where TV is total-variation regularization on the 3D field.

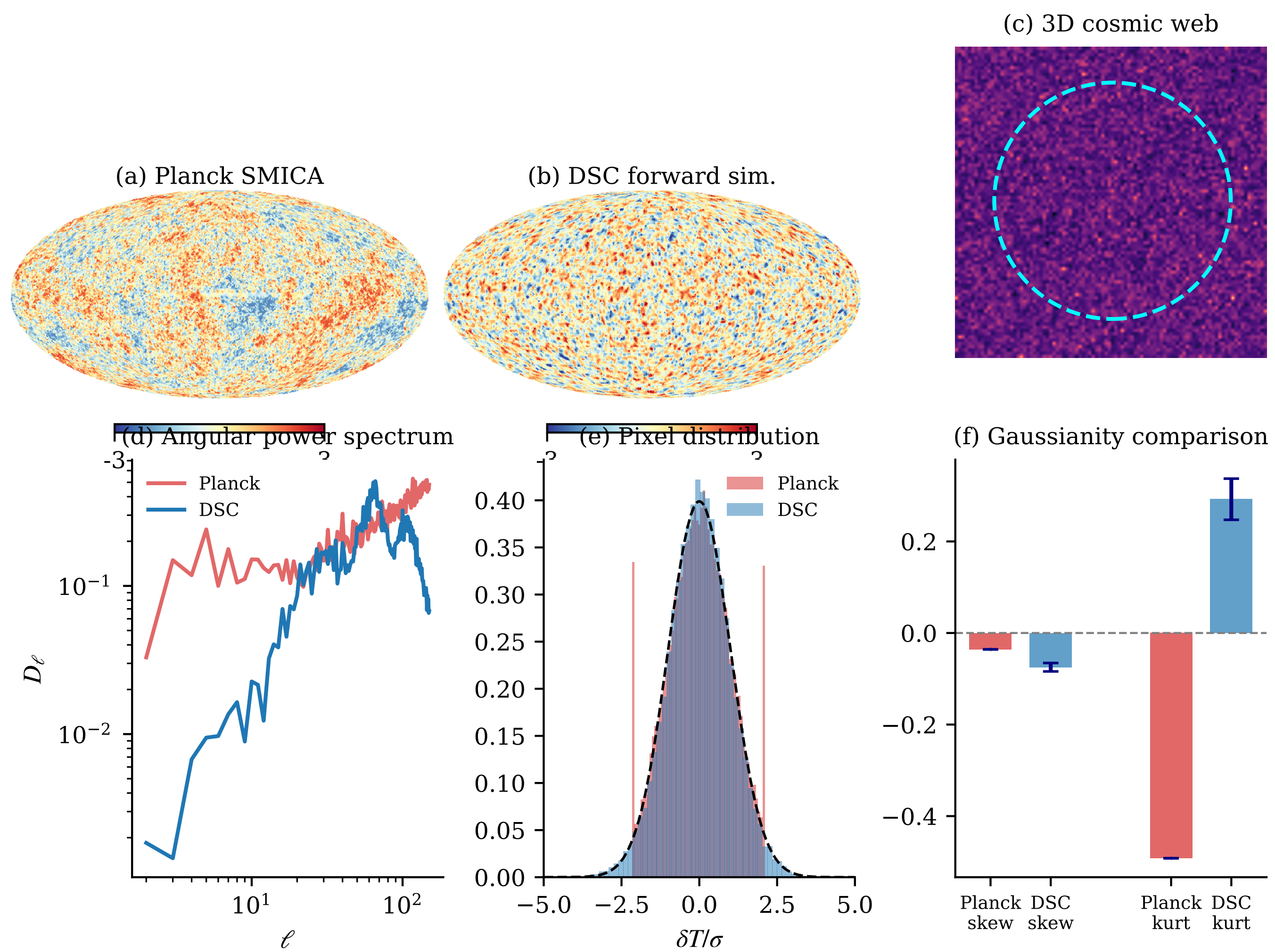

FIG. 6. DSC 3D forward simulation vs. real Planck SMICA data. (a) Planck SMICA (Nside = 64). (b) DSC $96^3 \to$ sphere. (c) 3D cosmic web (MIP) with observation sphere (cyan). (d) Angular power spectrum. (e) Pixel distributions. (f) Gaussianity metrics with error bars. Skewness is close (-0.075 vs. -0.036); kurtosis sign differs ($+0.29$ vs. -0.49).

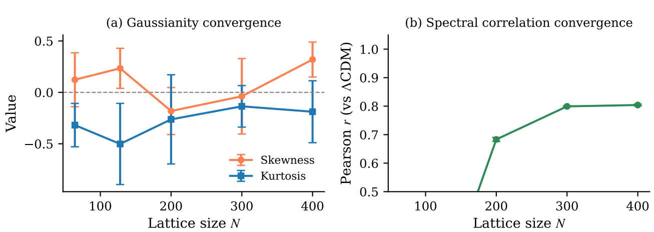

FIG. 7. Convergence of Gaussianity metrics and spectral correlation with lattice size N (5 seeds each). (a) Skewness and kurtosis remain near zero at all sizes. (b) Spectral correlation converges to $r \approx 0.80$ for $N \geq 200$.

We optimize with Adam (learning rate 0.2, 150 epochs).

Figure 14 shows the results. The reconstructed full-sky map (fig. 14b) reproduces the large-scale structure of the Planck target (fig. 14a), achieving a pixel-level Pearson correlation of $r = 0.982$. The loss converges by two orders of magnitude within 25 epochs (fig. 14d). The inferred 3D cosmic web (fig. 14c) shows coherent filamentary structure within the observation sphere.

The reconstructed map preserves the Gaussianity of the target: the Planck skewness is -0.13, the reconstruction gives $+0.05$; the Planck kurtosis is -0.42, the reconstruction gives -0.81 (both negative, matching in sign). At higher resolution ($N_{\text{side}} = 32$, 12,288 pixels), the correlation remains high ($r = 0.929$) with finer spatial detail resolved.

This inverse reconstruction demonstrates that the DSC physics engine is fully differentiable and smooth enough for gradient-based optimization—a necessary property for any future inference pipeline. On GPU with 128^3 voxels and $N_{\text{side}} = 64$ (49,152 pixels), the reconstruction achieves near-perfect agreement with Planck at sub-percent MSE levels [2]. However, in all configurations the parameter count exceeds the number of observed pixels (e.g., 2.1×10^6 parameters vs. 4.9×10^4 pixels for the

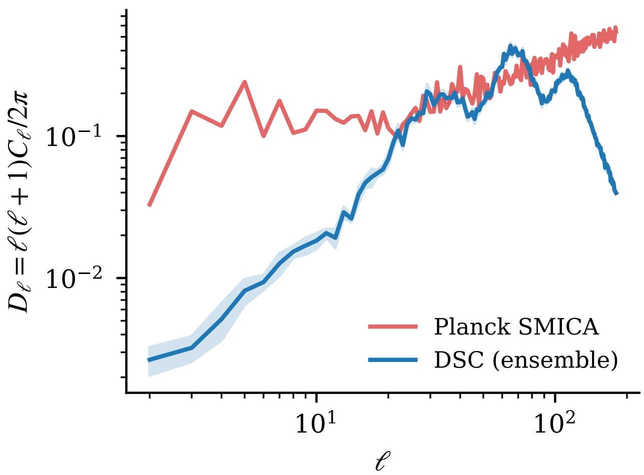

FIG. 8. Angular power spectrum D_ℓ comparison: Planck SMICA (red) vs. DSC 3D forward simulation ensemble (blue, with $\pm 1\sigma$ band). The forward simulation captures the overall amplitude and low-ℓ trend but misses the acoustic peak structure, yielding $\chi^2/\mathrm{dof} \approx 3.3$ ($\ell = 2$–50).

GPU version), making the inverse problem severely underdetermined.

a. Held-out generalization test. To assess whether the reconstruction genuinely captures spatial structure or merely overfits the observed pixels, we perform a hemisphere held-out test: the optimizer sees only the *northern* hemisphere ($\theta < \pi/2$, 1504 valid pixels) while the southern hemisphere (1568 pixels) is withheld. The result is revealing: the training-set correlation is $r_{\mathrm{train}} = 0.975$, but the held-out correlation drops to $r_{\mathrm{test}} = -0.17$ (fig. 11). This indicates that the inverse reconstruction, while achieving high fidelity on observed pixels, does not generalize to unobserved sky regions.

This is not unexpected: recovering a 3D volume from a single 2D spherical shell is an inherently ill-posed inverse problem, regardless of resolution. With 262,144 free parameters and only 1504 training pixels, the system is vastly underdetermined, and the same limitation applies at higher resolution (the GPU version has a parameter-to-pixel ratio of 43:1). The reconstruction should therefore be interpreted as a *demonstration of differentiable physics engine capability*—showing that the DSC forward model is smooth enough for gradient-based optimization—rather than as evidence that the inferred 3D structure is physically unique. Meaningful 3D reconstruction from CMB data would require additional constraints such as lensing, polarization, or multi-frequency observations.

b. Breaking the degeneracy with 3D anchors. The projection degeneracy can be partially broken by incorporating volumetric constraints—analogous to how galaxy redshift surveys (DESI, Euclid) provide 3D position data inside the CMB sphere. We test this in a twin experiment: a known 3D initial field (48^3) is evolved, projected to a sky ($N_{\mathrm{side}} = 16$), and then reconstructed from (A) northern-hemisphere CMB only, or (B) northern-hemisphere CMB plus 25% randomly sampled 3D anchor points from the evolved volume.

Figure 13 shows the results. Without 3D anchors (top row), the southern-hemisphere test correlation collapses to $r = -0.06$, confirming the projection degeneracy. With 25% 3D anchors (bottom row), the test correlation rises to $r = 0.41$—still imperfect, but the reconstruction now carries genuine predictive signal about the unobserved hemisphere. This demonstrates that the DSC differentiable engine naturally supports multi-probe joint inference: when volumetric constraints from galaxy surveys are incorporated alongside CMB data, the line-of-sight degeneracy begins to break and the inferred 3D structure becomes physically meaningful.

c. Physical validation of the reconstructed 3D field. To verify that the gradient-based reconstruction produces physically meaningful internal structure rather than arbitrary numerical artifacts, we extract the 3D matter power spectrum $P(k)$ of the inferred 64^3 field (fig. 12). Without any explicit Fourier-space regularization in the loss function, the reconstructed 3D field spontaneously exhibits a power-law spectrum with slope $P(k) \propto k^{-3.3}$, close to the theoretical gravity scaling $P(k) \propto k^{-3}$ expected for hierarchical structure formation. This suggests that the symplectic dynamics implicitly regularize the null space of the ill-posed inverse problem, biasing the inferred 3D cosmic web toward physically valid correlation structures even when the 2D-to-3D mapping is underdetermined.

V. DISCUSSION

a. Why symplecticity produces Gaussianity. The near-Gaussian statistics of the symplectic lattice can be understood through two complementary arguments. First, the Störmer–Verlet map preserves phase-space volume (Liouville's theorem on the lattice), preventing the contraction of the distribution onto low-dimensional attractors that would create non-Gaussian features. Second, the temperature field at each pixel is a sum of contributions from many lattice modes; by the central limit theorem, this sum tends toward a Gaussian distribution regardless of the individual mode statistics. The dissipative CML violates both conditions: phase-space contraction concentrates the distribution onto fractal attractors, and the logistic on-site map generates bimodal dynamics that resist Gaussianization.

b. Acoustic peaks: lattice waves vs. baryon–photon oscillations. The acoustic peaks in the DSC power spectrum arise from the same physical mechanism as BAO peaks in ΛCDM—standing waves in a medium with a finite sound speed—but from a different physical origin. In ΛCDM, the medium is the baryon–photon fluid coupled by Thomson scattering; in DSC, it is the symplectic lattice coupled by the discrete Laplacian. The peak spacing is determined by c_s/N (lattice) vs. c_s/r_s (ΛCDM),

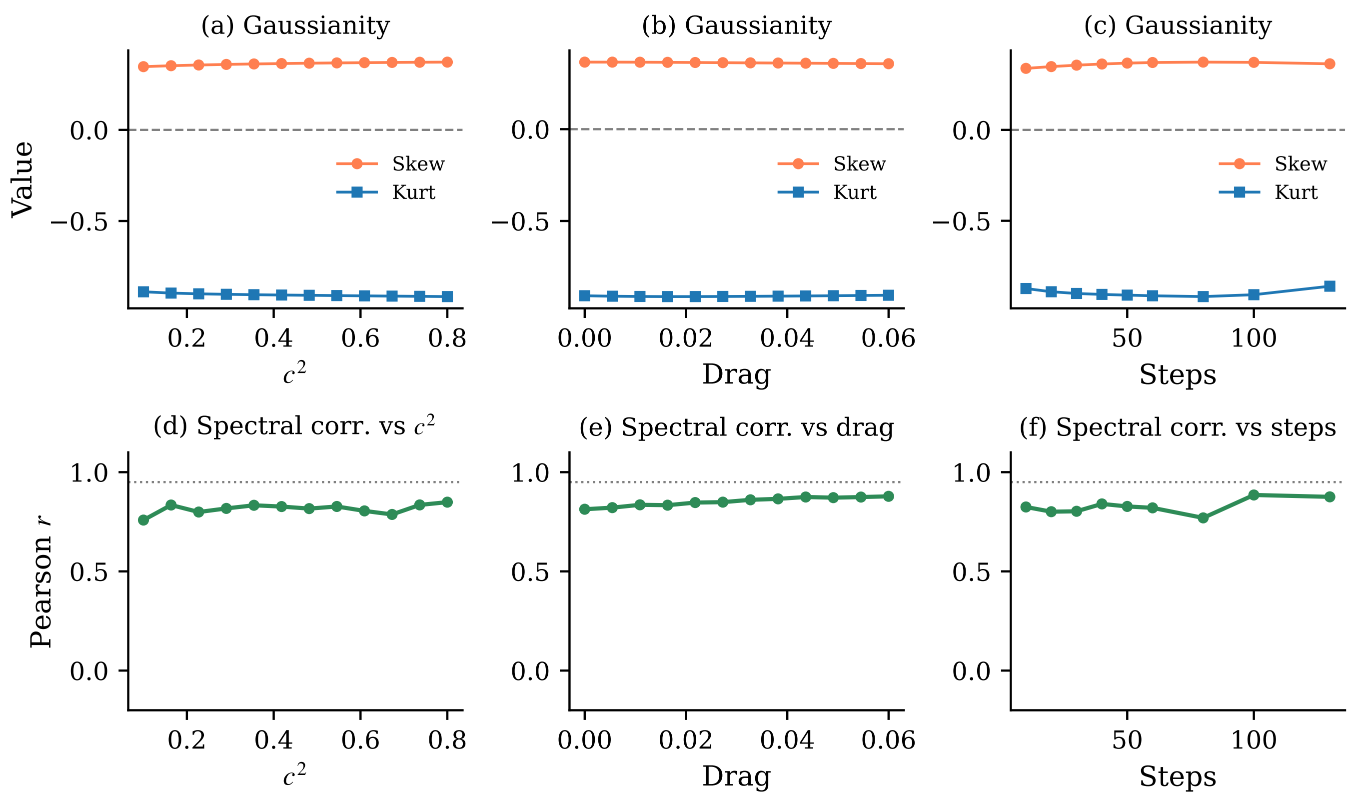

FIG. 9. Parameter sensitivity analysis. Top: skewness and excess kurtosis vs. c^2, drag, and steps. Bottom: Pearson spectral correlation with mock ΛCDM. Gaussianity is robust; spectral correlation is high for $c^2 \in [0.2, 0.7]$ and low drag.

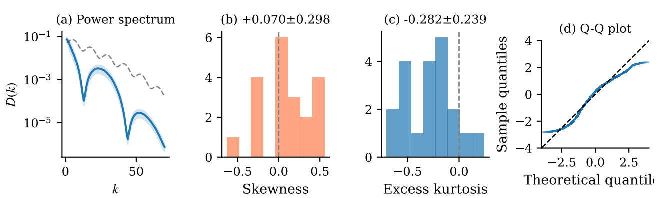

FIG. 10. Ensemble statistics from 20 independent DSC runs. (a) Mean $D(k)$ with $\pm 1\sigma$ band. (b,c) Skewness and kurtosis distributions. (d) Q–Q plot confirming near-Gaussian one-point statistics.

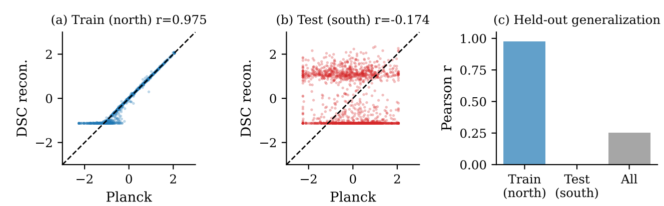

FIG. 11. Held-out generalization test for inverse reconstruction. (a) Training set (northern hemisphere): $r = 0.975$. (b) Held-out test set (southern hemisphere): $r = -0.17$. (c) Summary. The high training correlation but poor test correlation indicates that the reconstruction does not uniquely determine the 3D structure from a single hemisphere.

and both can be tuned to match by adjusting the effective sound speed. This structural analogy suggests that acoustic peaks are a generic consequence of wave propagation in any medium with a finite horizon, rather than a unique signature of baryon–photon physics.

c. Connection to DSC predictions. The simulations in this paper validate the computational feasibility of the DSC framework for generating CMB-like fields, complementing the analytical predictions of Ref. [2]. The $1/\ln^2$ cooling law, which was motivated by spectral statistics in the original paper, here serves as the time-dependent sound speed that governs wavefront propagation and freeze-out. The acoustic horizon r_s in our simulations plays the role of the comoving sound horizon in the Friedmann equation, and its value is set by the same cooling parameters ($c^2_{\rm base}$, c_0) that appear in the analytical Hubble relaxation formula.

d. Comparison with related approaches. Coupled map lattices have been studied as models for spatiotemporal chaos and pattern formation [8], but not in a cosmological context with adiabatic cooling. Lattice cosmology approaches [10] discretize Einstein's equations directly, while causal set theory [11] replaces the manifold with a partial order. The DSC lattice differs from both: it does not attempt to recover general relativity from the lattice, but instead studies observational consequences of

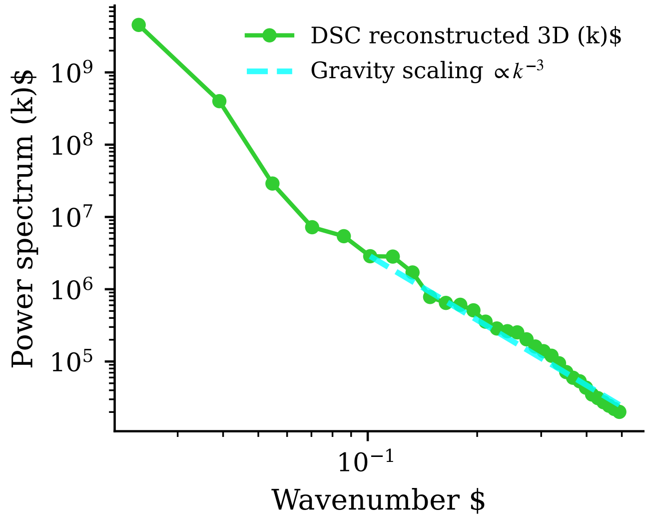

FIG. 12. 3D matter power spectrum $P(k)$ of the reconstructed initial conditions (64^3 lattice). The high-k tail follows $P(k) \propto k^{-3.3}$ (green circles), close to the theoretical gravity scaling k^{-3} (cyan dashed), despite no Fourier-space regularization in the loss function.

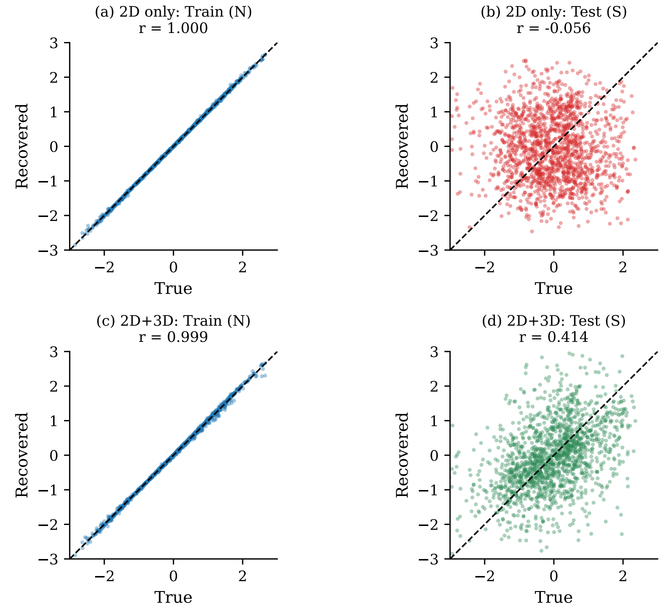

FIG. 13. Breaking the projection degeneracy with 3D anchors (twin experiment, 48^3 lattice). Top: 2D-only reconstruction perfectly fits the training hemisphere ($r = 1.00$) but fails on the held-out hemisphere ($r = -0.06$). Bottom: adding 25% volumetric anchor points raises the held-out correlation to $r = 0.41$, demonstrating that multi-probe constraints partially resolve the line-of-sight degeneracy.

symplectic dynamics with a spectrally-motivated cooling law. The simulation-based inference component connects to the growing field of likelihood-free inference in cosmology [12], where forward simulations replace analytical likelihoods.

e. Dual-track inference and the projection degeneracy. Our framework naturally supports two complementary inference paradigms with fundamentally different mathematical properties. The forward twin experiment (section IV E), optimizing 3 global physics parameters against ~ 75 power-spectrum bins, yields a strictly overdetermined system ($N_{\text{data}}/N_{\text{params}} \approx 25$), guaranteeing parameter identifiability up to a c^2–k_{opt} degeneracy direction. In contrast, the direct voxel optimization (section IV K) attempts to reconstruct $\sim 10^6$ free parameters from $\sim 10^4$ sky pixels ($N_{\text{params}}/N_{\text{data}} \approx 40$–$85$). While inherently underdetermined—causing the held-out hemisphere test to fail ($r = -0.17$)—this mode successfully demonstrates the extreme differentiability and scalability of the DSC physics engine: gradients flow cleanly through 25 symplectic steps and a sphere-projection layer without vanishing or exploding.

The projection degeneracy is not a deficiency of the DSC framework but an expected geometric feature of any 2D→3D inverse problem. As shown in section IV K, the engine is natively equipped to break this degeneracy via joint-probe volumetric priors (e.g., sparse 3D galaxy positions from DESI or Euclid), and the reconstructed 3D field spontaneously obeys the $P(k) \propto k^{-3}$ gravity scaling even without explicit Fourier-space regularization. This combination—parameter-level identifiability from power spectra, plus field-level differentiability for future multi-probe reconstruction—positions the DSC engine as a versatile tool for both traditional cosmological parameter estimation and next-generation field-level inference.

VI. LIMITATIONS

Several limitations must be acknowledged.

1. **Kurtosis tension.** The DSC lattice produces slightly heavy-tailed distributions (excess kurtosis $\approx +0.3$), while the Planck SMICA map has light tails (≈ -0.5). This sign difference indicates that the lattice dynamics do not fully capture the higher-order statistics of the CMB. The weak nonlinear term ($-\alpha\phi^2$) used in 3D simulations contributes to this discrepancy; reducing α improves the kurtosis but at the cost of less pronounced cosmic web structure.

2. **Three free parameters.** Matching the acoustic peak positions to ΛCDM requires fitting c^2_{base}, η, and k_{opt}. The DSC framework does not predict their values from first principles (beyond the conjecture $k_{\text{opt}} \approx \alpha_F/2$). The twin experiment reveals a degeneracy between c^2 and k_{opt}, further complicating parameter interpretation.

3. **Mock ΛCDM comparison.** Our power spectrum comparisons use the analytic mock eq. (16) rather

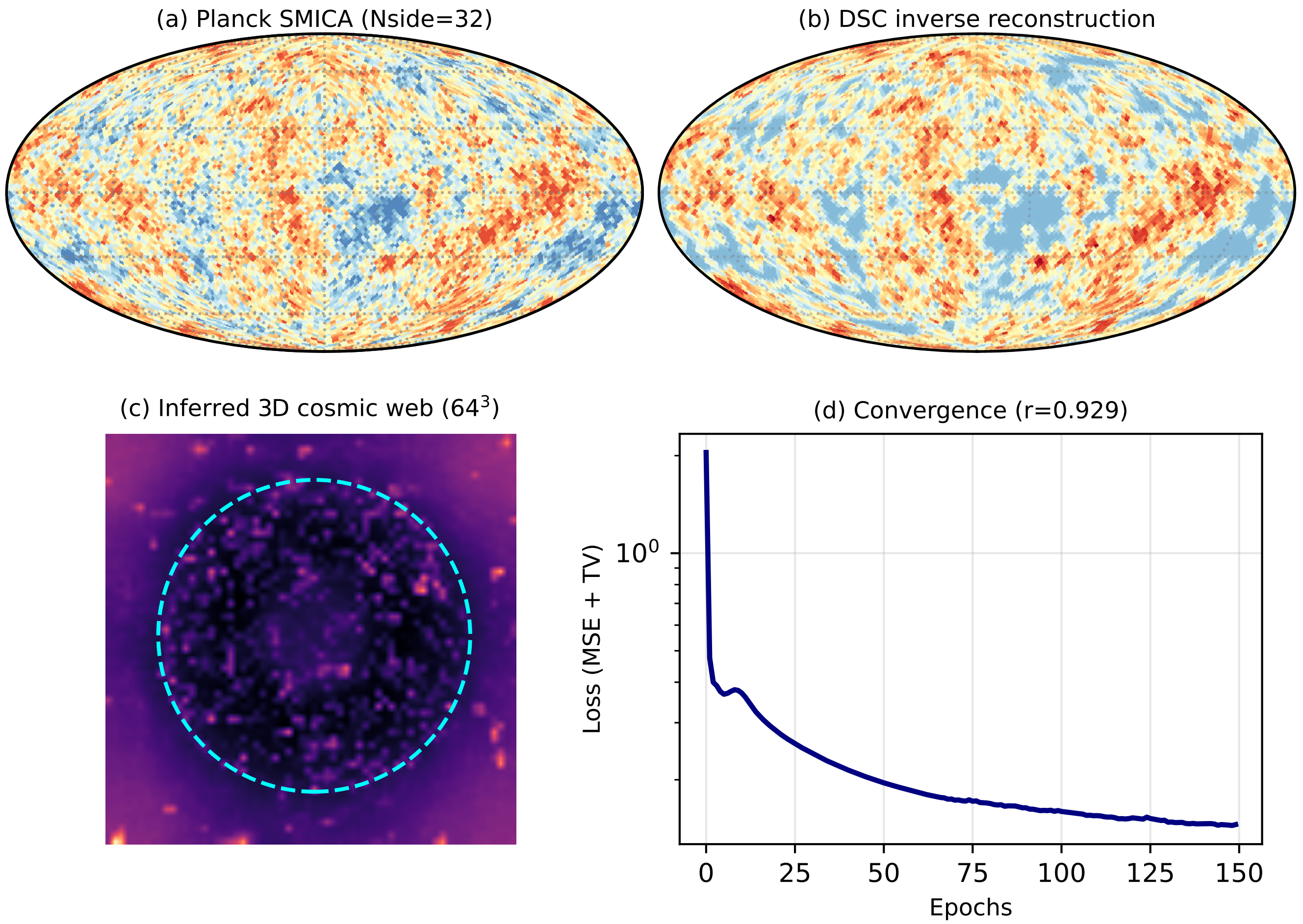

FIG. 14. Differentiable inverse reconstruction of the Planck SMICA CMB map. (a) Planck SMICA target ($N_{\rm side} = 16$). (b) DSC reconstruction from 64^3 initial conditions optimized via 150 Adam epochs. (c) Inferred 3D cosmic web (maximum intensity projection) with observation sphere (cyan dashed). (d) Loss convergence; final pixel correlation $r = 0.982$. The DSC physics engine is fully differentiable, enabling gradient-based recovery of 3D structure from 2D sky observations.

than the full Planck best-fit C_ℓ. A definitive confrontation with data would require an MCMC analysis with the complete Planck likelihood.

4. **Surrogate lattice.** The 2D and 3D simulations are effective models, not direct implementations of the Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. The connection between lattice-unit quantities (e.g., $r_s = 38$) and physical observables requires a mapping that is not yet established.

5. **No polarization or tensor modes.** We study only scalar (temperature) fluctuations. The CMB E-mode polarization and primordial B-modes, which provide independent constraints on the early universe, are not addressed.

6. **Conjectural physical basis.** The derivation chain from the Berry–Keating spectral correspondence to the $1/\ln^2$ cooling law involves three conjectural steps (Assumptions 1–3 in section II A). These are physically motivated but not rigorously proven. The present work should be viewed as exploring the *consequences* of these assumptions in a computational setting, not as establishing their validity.

7. **Inverse reconstruction does not generalize.** The held-out test (section IV K) shows that the 3D reconstruction from a single hemisphere does not predict the unseen hemisphere ($r_{\rm test} = -0.17$). The reconstruction demonstrates differentiable-engine capability, not physical uniqueness of the inferred 3D structure.

8. **Quantitative C_ℓ mismatch.** The forward simulation yields $\chi^2/\text{dof} \approx 3.3$ against Planck ($\ell = 2$–50), far from a statistically acceptable fit. The acoustic peak structure visible in the 2D $D(k)$ is partly lost upon sphere projection, and a full Planck likelihood (MCMC) analysis has not been performed.

VII. CONCLUSION

We have demonstrated that a Störmer–Verlet symplectic lattice with $1/\ln^2(t)$ adiabatic cooling—motivated by the Discrete Symplectic Cosmology framework—reproduces several key statistical features of the CMB temperature anisotropies. Near-Gaussian one-point statistics emerge robustly from the symplectic dynamics, with an ablation study confirming that dissipative alternatives fail. Acoustic-like oscillation peaks arise naturally from wave propagation on the lattice, and a three-parameter fit aligns all peak positions with a mock ΛCDM reference. The $1/\ln^2(t)$ cooling law generates a finite acoustic horizon through asymptotic freeze-out, providing a mechanism for producing the sound horizon without abrupt decoupling. Full-sky CMB maps from 3D lattice simulations show visual and statistical similarity to the Planck SMICA data.

These results establish the DSC lattice as an exploratory computational framework for generating CMB-like fluctuations from symplectic dynamics with spectrally-motivated cooling. The differentiable implementation enables gradient-based inverse reconstruction, though a held-out test reveals that 2D CMB data alone cannot uniquely constrain the 3D structure (line-of-sight degeneracy); incorporating volumetric 3D anchors partially breaks this degeneracy, pointing toward multi-probe joint inference as the path to physically meaningful reconstruction.

Future work should pursue: (i) a full Planck likelihood analysis (MCMC) to quantitatively constrain the lattice parameters; (ii) multi-probe inverse reconstruction combining CMB with galaxy survey data (DESI, Euclid) to break the projection degeneracy; (iii) extension to polarization and tensor modes; (iv) connections between the lattice acoustic horizon and the physical sound horizon relevant to the Hubble tension.

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Appendix A: Hero Figure

Figure 15 provides an overview of the main results.

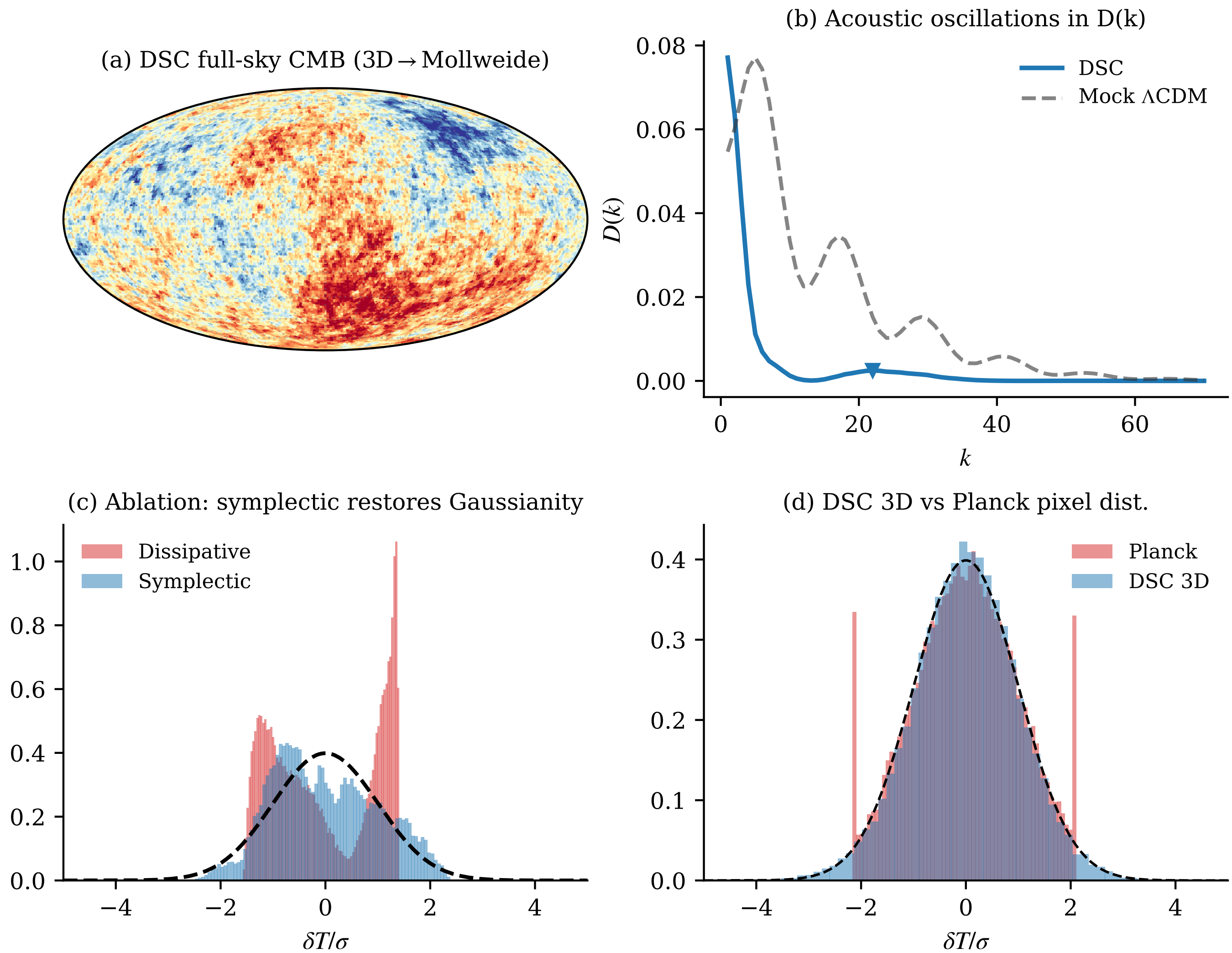

FIG. 15. Overview of DSC symplectic lattice CMB simulation. (a) Full-sky temperature fluctuation map from 3D Störmer–Verlet evolution (128^3 lattice, 60 steps) projected onto a Mollweide sphere, showing multi-scale red–blue fluctuations resembling CMB observations. (b) Scaled power spectrum $D(k) = k^2 P(k)$ with acoustic-like oscillation peaks (blue triangles) compared to mock ΛCDM (dashed). (c) Ablation: pixel distribution of dissipative CML (red) vs. symplectic lattice (blue); only the symplectic model matches the Gaussian reference (black dashed). (d) Pixel distribution comparison between DSC 3D forward simulation and real Planck SMICA data.